

INDIA DATA RESEARCH ACADEMY
DATA SCIENCE & AI TRAINING PROGRAM

THE IMPACT OF SOCIO-ECONOMIC FACTORS ON ACADEMIC PERFORMANCE

A Comprehensive Analysis of Student Outcomes

Regression-Based Analysis and Prediction of Exam Score

Capstone Project Report Submitted in Fulfilment of the Requirements
for the Data Science & AI Training Program by IDRA

Department of Data Science and AI

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DECLARATION

I, Abhinav Jacob Sunny, hereby declare that the work presented in this capstone project report, titled “The Impact of Socio-Economic Factors on Academic Performance: A Comprehensive Analysis of Student Outcomes,” is my own original work, carried out as part of the Data Science & AI Training Program conducted by the India Data Research Academy (IDRA).

All data analysis, coding, and modelling presented in this report and in the accompanying Jupyter/Colab notebook were completed by me. Wherever external datasets, documentation, or reference material have been used, they have been duly acknowledged in the References section. This report has not been submitted, in whole or in part, for any other academic award or qualification.

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ABSTRACT

This project investigates the factors associated with student academic performance using the supplied Student Performance Factors dataset. The original dataset contains 6,607 observations and 20 variables, comprising 7 numerical and 13 categorical fields, with Exam_Score as the prediction target. Data-quality inspection found no duplicate records, missing values in Teacher_Quality, Parental_Education_Level, and Distance_from_Home, and one invalid Exam_Score value (101) outside the logical 0–100 range, which was excluded prior to analysis. Remaining categorical missingness was handled through most-frequent imputation and numerical features through median imputation and standard scaling, both fitted within the training workflow to avoid data leakage.

Exploratory analysis examined distributions, correlations, and group-level differences across socio-economic and behavioural variables. Attendance showed the strongest numerical correlation with Exam_Score ($r \approx 0.581$), followed by Hours_Studied ($r \approx 0.445$), while parental involvement, access to resources, and peer influence showed smaller but consistent group-level differences. Three regression models — Linear Regression, Random Forest, and Gradient Boosting — were trained on an 80/20 train-test split and compared using MAE, RMSE, and R^2 . Linear Regression generalised best on the held-out test set, achieving a test RMSE of 1.521, MAE of 0.416, and R^2 of 0.711, while Random Forest showed a substantially larger train-test performance gap indicative of overfitting.

The findings suggest that attendance and study effort are the most useful behavioural predictors of exam performance in this dataset, while socio-economic and learning-environment variables provide meaningful, though comparatively smaller, contextual signal. The results are best interpreted as predictive and associative rather than causal.

Keywords: EDA; Data Preprocessing; Student Performance; Socio-Economic Factors; Regression; Predictive Analytics

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1. INTRODUCTION

1.1 Background

Academic performance is shaped by a combination of individual study behaviour, family background, and the quality of the learning environment surrounding a student. Understanding how these socio-economic and behavioural factors relate to examination outcomes can help educators and institutions identify where support is most likely to be effective. This project examines a dataset of student records capturing study habits, attendance, parental involvement, access to educational resources, and other contextual variables, with the aim of understanding and predicting exam performance (Exam_Score).

1.2 Context of the Problem

In real classrooms, teachers and administrators often have access to background information about students — such as attendance records, parental involvement, and resource access — well before final examination results are available. If certain factors are reliably associated with performance, this information could support early identification of students who may benefit from additional academic support. This project is framed from that practical perspective: using available student-level data to understand and forecast exam performance.

1.3 Scope of the Study

This study is limited to the supplied Student Performance Factors dataset (6,607 records, 20 variables) and does not draw on additional external data sources. The analysis covers data cleaning, exploratory data analysis, statistical description, and regression modelling of Exam_Score. The study does not attempt to establish causal relationships between socio-economic factors and performance; it reports descriptive and predictive associations only. Model deployment and real-time application are outside the scope of this project.

2. PROBLEM STATEMENT AND MOTIVATION

Given the wide range of behavioural and contextual factors that could influence a student's exam performance, it is not immediately obvious which variables carry the most predictive value, or how accurately performance can be forecast using observable indicators such as attendance and study hours.

This project addresses the following problem: which socio-economic and behavioural factors are most strongly associated with student exam performance, and how accurately can Exam_Score be predicted using these factors? A data-driven answer to this question can help identify practical, low-cost indicators that institutions could monitor to support students proactively, rather than relying on assumption or anecdote.

3. OBJECTIVES

- To understand and prepare the Student Performance Factors dataset for analysis, including identifying and resolving data-quality issues.
- To identify missing values, duplicate records, and invalid target values, and to treat them appropriately.
- To explore numerical relationships and categorical group differences associated with Exam_Score.
- To build and compare multiple regression models capable of predicting Exam_Score.
- To evaluate model generalisation using a held-out test set and appropriate regression metrics.

- To translate the analytical and modelling results into evidence-based findings, limitations, and recommendations.

4. RESEARCH QUESTIONS

1. What are the major characteristics and data-quality issues present in the Student Performance Factors dataset?
2. Which numerical and categorical variables are most strongly associated with Exam_Score?
3. How does Exam_Score vary across parental involvement, access to resources, and peer influence groups?
4. How accurately can Exam_Score be predicted using the available student characteristics, and which regression model generalises best?

Primary research question: *How well can student characteristics and learning-related factors predict Exam_Score?*

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used in this project, P_2_StudentPerformanceFactors.csv, was supplied for the IDRA Data Science & AI Capstone Project as the dataset corresponding to Project Title 2: “The Impact of Socio-Economic Factors on Academic Performance.” No additional external data was merged with the supplied file.

5.2 Dataset Size and Structure

The dataset contains 6,607 rows and 20 columns, comprising 7 numerical and 13 categorical variables. The prediction target is Exam_Score. The supplied fields cover study behaviour, attendance, parental involvement, access to resources, motivation, family income, teacher quality, school type, peer influence, and other student characteristics.

Table 1. Summary of the Dataset

Metric	Value
Number of rows (original)	6,607
Number of columns	20
Numerical variables	7
Categorical variables	13
Target variable	Exam_Score
Duplicate rows found	0
Rows after cleaning	6,606

5.3 Data Dictionary

Table 2 summarises the role of each variable in the dataset.

Table 2. Data Dictionary

Variable	Type	Role / Description
Hours_Studied	Numerical	Weekly study effort (hours)
Attendance	Numerical	Attendance percentage
Parental_Involvement	Categorical	Level of parental involvement (Low/Medium/High)
Access_to_Resources	Categorical	Access to educational resources
Extracurricular_Activities	Categorical	Participation in extracurricular activities
Sleep_Hours	Numerical	Average sleep duration
Previous_Scores	Numerical	Prior academic score
Motivation_Level	Categorical	Self-reported motivation category
Internet_Access	Categorical	Availability of internet access
Tutoring_Sessions	Numerical	Number of tutoring sessions attended
Family_Income	Categorical	Family income category
Teacher_Quality	Categorical	Perceived teacher quality
School_Type	Categorical	Type of school attended
Peer_Influence	Categorical	Nature of peer influence (Positive/Neutral/Negative)
Physical_Activity	Numerical	Physical activity level
Learning_Disabilities	Categorical	Presence of a learning disability
Parental_Education_Level	Categorical	Highest parental education level
Distance_from_Home	Categorical	Distance category from home to school
Gender	Categorical	Student gender
Exam_Score	Numerical	Prediction target

6. DATA CLEANING

This section documents the state of the raw data prior to cleaning and the treatment applied to each identified issue.

6.1 Missing Values

Missing values were found only in three categorical columns, as summarised below.

Column	Missing Count	% of Records
Teacher_Quality	78	1.18%
Parental_Education_Level	90	1.36%
Distance_from_Home	67	1.01%

Because the affected proportion of records is small (under 1.5% for each column) and the missing fields are categorical, rows were not deleted. Instead, missing values were handled through most-frequent (mode) imputation applied within the modelling pipeline, fitted on the training fold only. This preserves sample size and avoids the bias that could result from dropping a large number of otherwise complete records.

6.2 Duplicate Records

A full-row duplicate check found 0 duplicate records in the supplied dataset. No rows were removed for this reason.

6.3 Data Types and Inconsistent Values

All numerical columns were confirmed to hold numeric data types after loading. Categorical columns were checked for inconsistent capitalisation or category labels (for example, “low” versus “Low”); no inconsistent category values were found, and no further text-normalisation was required.

6.4 Outlier and Invalid Value Handling

One record contained an Exam_Score value of 101, which lies outside the logical 0–100 scoring range and therefore represents a data-entry error rather than a legitimate extreme value. This record was excluded, leaving 6,606 observations. Other numerical variables (such as Hours_Studied and Tutoring_Sessions) were checked for extreme values using range and distribution inspection; the values observed were considered plausible for a student population and were retained rather than removed, since trimming them would not be justified by any logical constraint.

6.5 Before vs After Cleaning

Table 3. Data Quality Before and After Cleaning

Metric	Before Cleaning	After Cleaning
Total rows	6,607	6,606
Duplicate rows	0	0
Invalid Exam_Score values (>100)	1	0
Missing — Teacher_Quality	78	0 (imputed in pipeline)
Missing — Parental_Education_Level	90	0 (imputed in pipeline)
Missing — Distance_from_Home	67	0 (imputed in pipeline)

7. DATA PREPROCESSING

7.1 Categorical Encoding

All 13 categorical variables were encoded using one-hot encoding, converting each category into a binary indicator column. One-hot encoding was chosen over ordinal encoding because most categorical fields (e.g. School_Type, Gender, Internet_Access) do not have a natural order, and applying a uniform encoding scheme keeps the preprocessing pipeline simple and consistent across all categorical inputs.

7.2 Feature Scaling and Transformation

Numerical features were standardised (zero mean, unit variance). Standardisation is particularly relevant for Linear Regression and keeps all numerical inputs on a comparable scale for fair comparison across the three candidate models.

7.3 Train-Test Split

The cleaned dataset (6,606 records) was divided into 80% training data and 20% testing data using random_state=42 for reproducibility. All imputation, encoding, and scaling transformations were fitted

exclusively on the training fold and then applied to the test fold, preventing information from the test set from influencing preprocessing — a safeguard against data leakage.

7.4 Final Feature Set

The final feature set (X) comprises the 19 non-target variables listed in Table 2 (7 numerical, 13 categorical, expanded via one-hot encoding). The target variable (y) is Exam_Score.

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

Figure 1 shows the distribution of Exam_Score after removing the invalid record. The distribution is unimodal and right-skewed, with the large majority of students scoring between 60 and 75, and a small number of higher-scoring outliers extending up to 100.

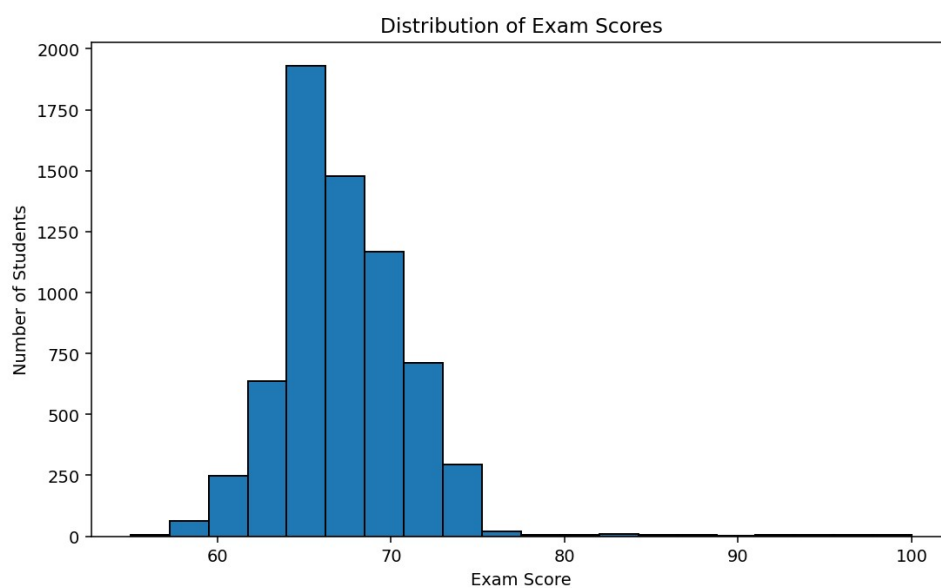


Figure 1. Distribution of Exam Scores.

8.2 Bivariate / Relationship Analysis

Figures 2 and 3 examine the relationship between Exam_Score and the two variables identified as most strongly correlated with it: Attendance and Hours_Studied.

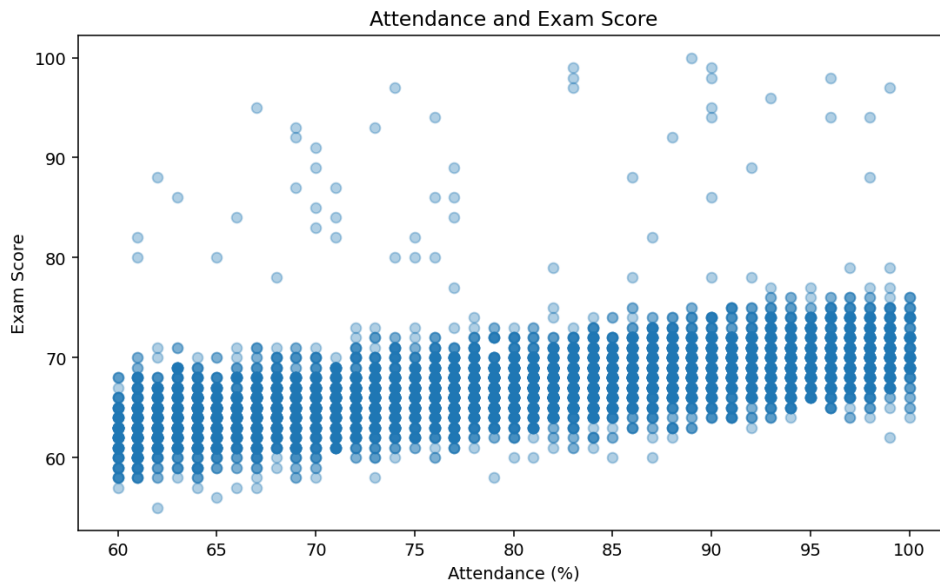


Figure 2. Attendance and Exam Score.

Figure 2 shows a positive association between attendance percentage and exam score: students with higher attendance tend to cluster toward higher scores, while the small number of high-scoring outliers at low attendance are comparatively rare.

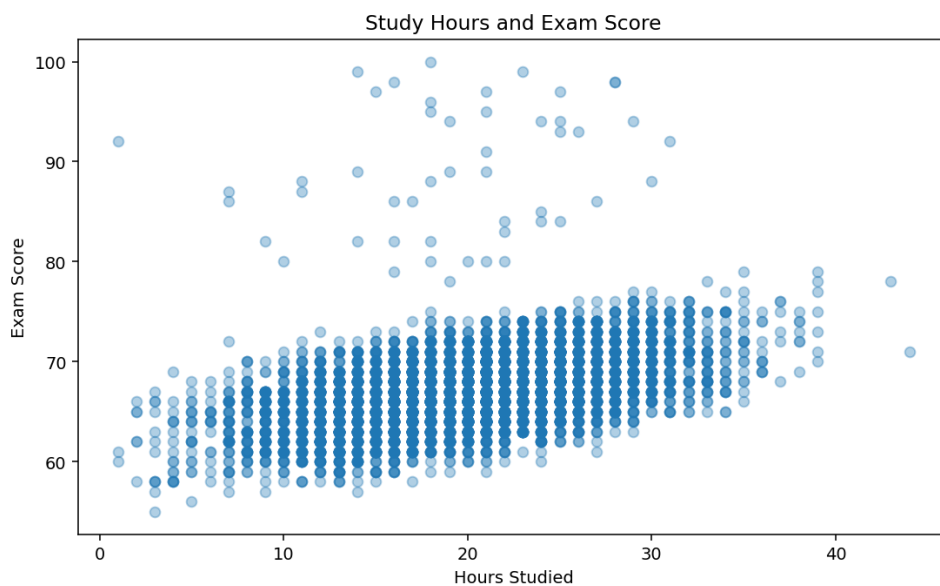


Figure 3. Study Hours and Exam Score.

Figure 3 shows a similar positive relationship between hours studied and exam score, though with more spread than attendance, particularly among students who report low study hours but achieve unusually high scores.

8.3 Correlation Analysis

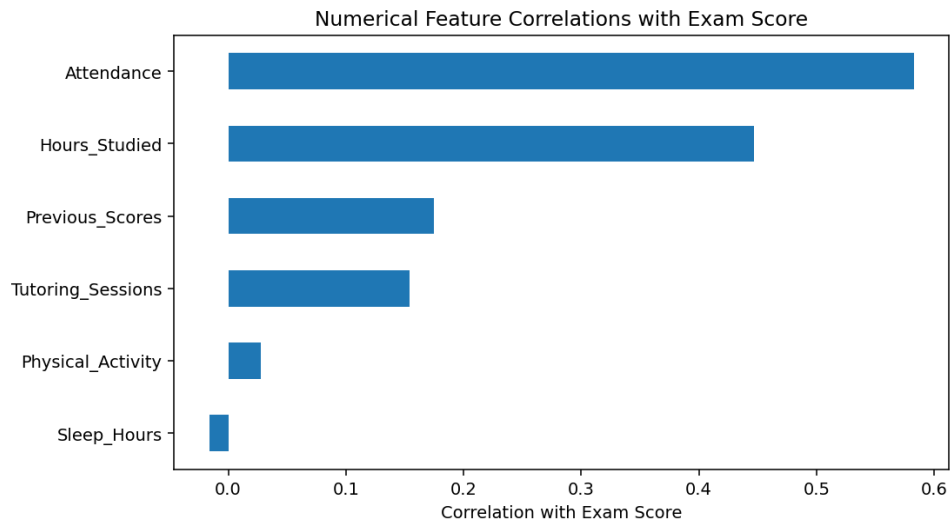


Figure 4. Numerical Correlations with Exam Score.

Attendance has the strongest numerical association with Exam_Score ($r \approx 0.581$), followed by Hours_Studied ($r \approx 0.445$). Previous_Scores and Tutoring_Sessions show weaker positive associations, while Sleep_Hours and Physical_Activity show relatively small linear correlations with the target. These results guided feature interpretation during modelling: attendance and study effort emerged as the numerical variables most worth emphasising in the discussion of results.

8.4 Categorical / Group Comparison Analysis

Figures 5 to 7 compare mean Exam_Score across three contextual, socio-economic and environment-related categorical variables.

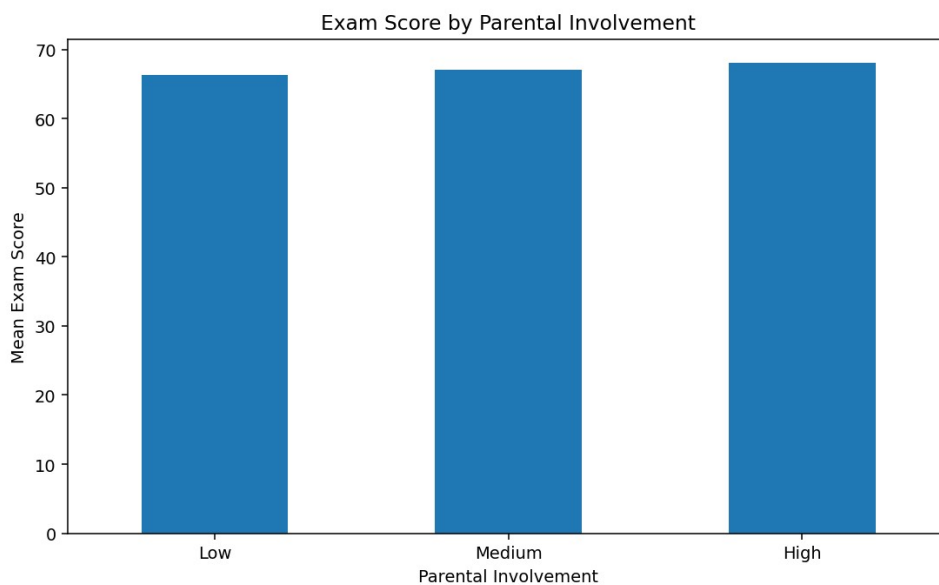


Figure 5. Exam Score by Parental Involvement.

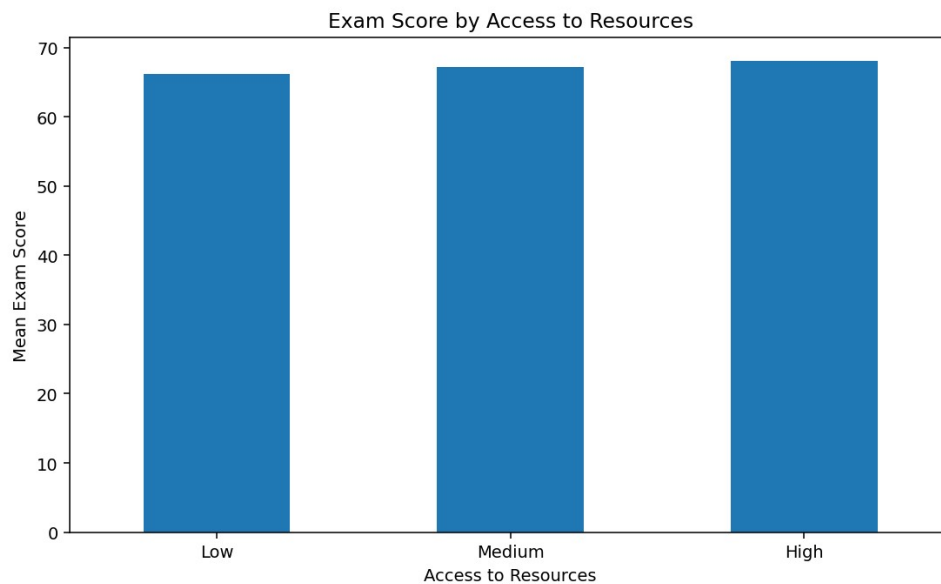


Figure 6. Exam Score by Access to Resources.

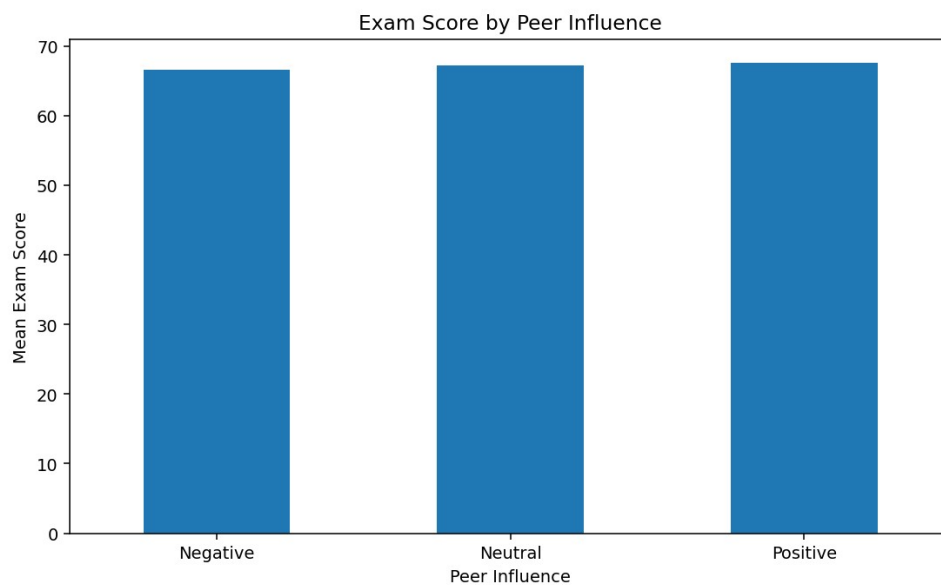


Figure 7. Exam Score by Peer Influence.

Group comparisons show consistently higher average scores for students with high parental involvement, high access to resources, and positive peer influence, compared with their lower or negative counterparts. The differences between groups are smaller than the numerical relationships shown in Figure 4, but they are directionally consistent across all three variables. These are descriptive, associative differences and do not by themselves establish causation.

9. STATISTICAL ANALYSIS

Table 4 reports descriptive statistics for the target variable, Exam_Score, after cleaning.

Table 4. Descriptive Statistics of Exam Score

Statistic	Value
Mean	67.23
Median	67.00

Statistic	Value
Standard Deviation	3.87
Minimum	55
Maximum	100
Range	45

The mean and median are close in value (67.23 versus 67.00), but the maximum (100) sits far beyond the third quartile relative to the standard deviation, which is consistent with the right-skewed shape observed in Figure 1: most students score within a narrow band, while a small number of high-performing students pull the upper tail of the distribution outward. This supports the use of robust central-tendency reporting (median alongside mean) when summarising typical performance in this dataset.

10. FEATURE ENGINEERING

No additional derived features were engineered for this project. The supplied dataset already provides sufficiently granular, directly interpretable variables covering study behaviour (Hours_Studied, Tutoring_Sessions), attendance, prior performance (Previous_Scores), and socio-economic/environmental context (Parental_Involvement, Access_to_Resources, Family_Income, Parental_Education_Level, Distance_from_Home, Peer_Influence). Given this, feature preparation was limited to the standard preprocessing transformations described in Section 7 (encoding and scaling) rather than the creation of new composite variables, since no clear domain-justified interaction or ratio variable was identified that would add information beyond what the existing fields already capture.

11. MODEL DEVELOPMENT

11.1 Problem Formulation

The task is formulated as a regression problem because Exam_Score is a continuous numerical variable. Features (X) and target (y) were separated prior to any preprocessing or modelling step, and the 19 input variables described in Section 7.4 were used to predict Exam_Score.

11.2 Model Selection

Three regression models were selected to represent a range of modelling assumptions:

- Linear Regression — a simple, interpretable baseline that assumes an approximately linear relationship between features and the target.
- Random Forest Regressor — an ensemble of decision trees capable of capturing non-linear relationships and interactions between variables.
- Gradient Boosting Regressor — a sequential boosting ensemble that iteratively corrects prediction errors, often effective on tabular data.

Comparing these three allows the project to assess whether the additional complexity of tree-based ensembles improves prediction over a simple linear baseline for this particular dataset.

11.3 Training the Model

All models were trained on the same 80/20 train-test split (random_state=42), using the preprocessing pipeline described in Section 7. An illustrative extract of the training code is shown below; the complete, executable pipeline is provided in the accompanying notebook and in Appendix A.

```

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42)

numeric_pipe = Pipeline([('impute', SimpleImputer(strategy='median')),
                          ('scale', StandardScaler())])
categorical_pipe = Pipeline([('impute', SimpleImputer(strategy='most_frequent')),
                              ('encode', OneHotEncoder(handle_unknown='ignore'))])
preprocessor = ColumnTransformer([('num', numeric_pipe, numeric_cols),
                                  ('cat', categorical_pipe, categorical_cols)])

model = Pipeline([('preprocess', preprocessor),
                  ('regressor', LinearRegression())])
model.fit(X_train, y_train)

```

12. MODEL EVALUATION

12.1 Regression Evaluation

Model performance was evaluated on the held-out test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . Lower MAE/RMSE indicate lower prediction error, while a higher R^2 indicates a greater proportion of variance in Exam_Score explained by the model.

Table 5. Regression Model Performance

Model	Train RMSE	Test RMSE	Test MAE	Test R^2
Linear Regression	2.109	1.521	0.416	0.711
Random Forest	1.379	1.912	1.028	0.877
Gradient Boosting	2.105	1.764	0.850	0.712

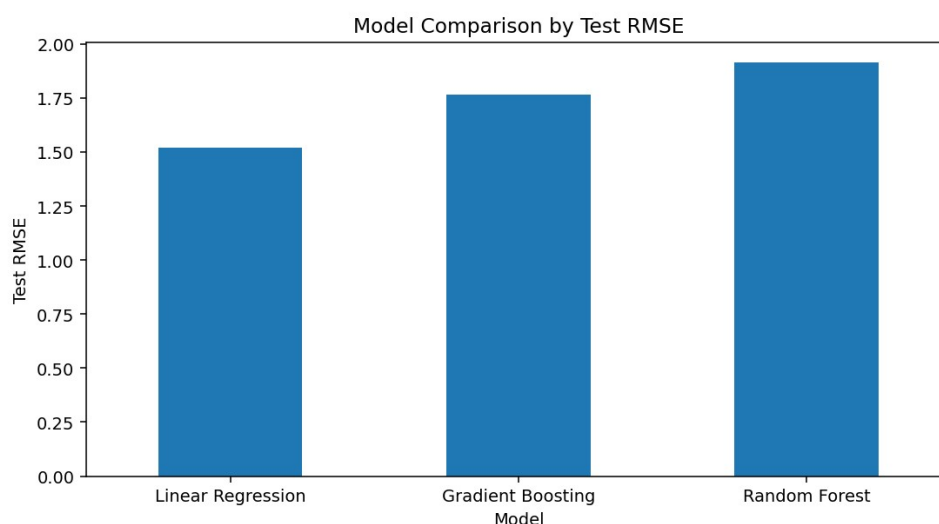


Figure 8. Model Comparison by Test RMSE.

Selected model: Linear Regression. It achieved the lowest test RMSE (1.521) and test MAE (0.416) among the three models evaluated. Although Random Forest reported a higher test R^2 (0.877), its test RMSE and MAE were both worse than Linear Regression's, and its much lower training RMSE (1.379) relative to its test

RMSE (1.912) points to overfitting rather than genuinely superior generalisation. Linear Regression was therefore selected as the final model on the basis of its more consistent train-test performance.

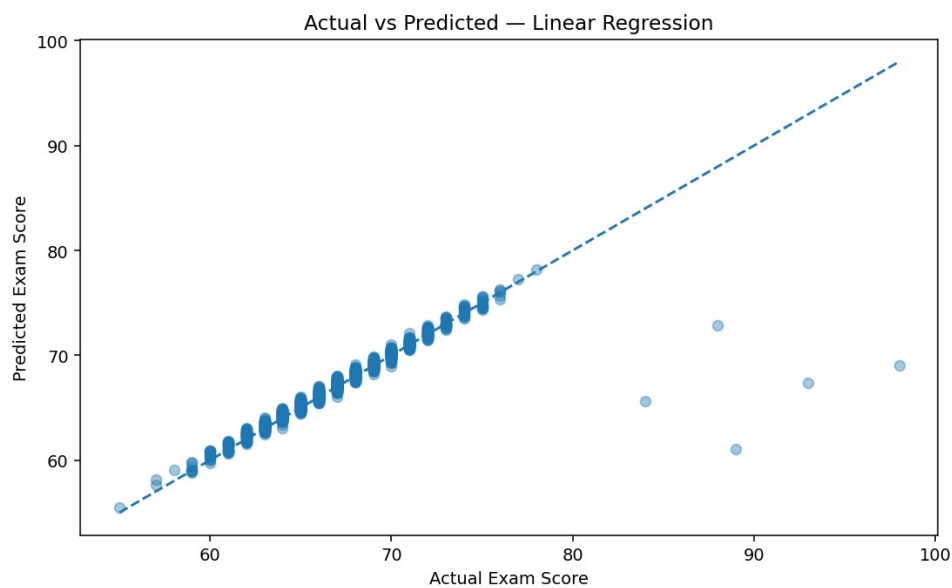


Figure 9. Actual versus Predicted Scores — Selected Model.

Figure 9 shows that predictions track the actual scores closely for the bulk of the distribution, with residuals widening for the small number of students who scored well above the typical 60–75 range — the model tends to underestimate these exceptionally high performers.

12.2 Overfitting and Underfitting

Random Forest shows a substantially larger gap between its training RMSE (1.379) and test RMSE (1.912) than Linear Regression (2.109 versus 1.521), indicating that it fits the training data more closely than it generalises to unseen data — a signature of overfitting. Gradient Boosting shows a smaller train-test gap than Random Forest but still underperforms Linear Regression on test RMSE and MAE. Linear Regression shows the most consistent train-test performance of the three, and generalises best on this held-out split, supporting its selection as the final model.

13. RESULTS AND FINDINGS

Finding 1: Attendance is the strongest single predictor of Exam_Score ($r \approx 0.581$, Figure 4 and Figure 2). Practically, attendance is a low-cost, easily monitored indicator that institutions could track to flag students who may need support.

Finding 2: Study effort (Hours_Studied) is also strongly and positively associated with Exam_Score ($r \approx 0.445$, Figure 3), supporting the value of encouraging structured, consistent study routines over sporadic effort.

Finding 3: Contextual and socio-economic variables — parental involvement, access to resources, and peer influence (Figures 5–7) — show smaller but directionally consistent positive differences between higher and lower categories, suggesting environment plays a supporting, though secondary, role relative to individual behaviour.

Finding 4: Linear Regression generalised best among the three models tested (Table 5, Figure 8), with test RMSE = 1.521, MAE = 0.416, and $R^2 = 0.711$, while Random Forest's larger train-test gap indicates overfitting despite its higher raw R^2 .

Finding 5: Prediction accuracy is strongest for the majority of students clustered in the 60–75 score range, but the model under-predicts a small number of exceptionally high-scoring students (Figure 9), indicating a limitation in modelling the upper tail of performance.

14. DISCUSSION

The EDA results and the model evaluation tell a consistent story: attendance and study effort are the most prominent numerical predictors of Exam_Score in this dataset, while socio-economic and learning-environment variables contribute meaningful but comparatively smaller group-level differences. This pattern is intuitive — attendance and study time are direct, proximate behaviours linked to exam preparation, whereas parental involvement, resource access, and peer influence are more distal factors that likely act through their effect on those proximate behaviours.

The selected Linear Regression model provides a useful predictive baseline, achieving a moderate-to-strong R^2 (0.711) with low absolute error (MAE $\approx$ 0.42 points on a roughly 45-point score range). However, the comparison with Random Forest illustrates an important methodological point: a model with a higher R^2 is not automatically the better choice if it also shows a much larger train-test performance gap. Model selection in this project therefore prioritised generalisation over raw explained variance.

Overall, the results should be interpreted as predictive and associative rather than causal. The dataset does not support claims that, for example, increasing parental involvement directly causes higher exam scores; it only supports the observation that these variables move together in the available sample.

15. LIMITATIONS

- The dataset may omit important influences on academic performance (e.g. detailed socio-economic indicators, mental health, or classroom-level factors) not captured in the supplied fields.
- The sample may not represent every student population, school system, or academic context, which limits generalisation beyond the dataset's source population.
- Missing categorical values required imputation, which introduces some uncertainty into a small proportion of records (under 1.5% per affected column).
- Results depend on the specific train-test split used; a different random split could yield somewhat different metric values.
- The analysis identifies associations and predictive relationships, not causal effects — correlation between variables such as parental involvement and Exam_Score does not establish that one causes the other.
- Only three regression model families were tested; more extensive hyperparameter tuning or additional model types might yield further improvements.

16. CONCLUSION

This study completed a full data-science workflow — from data-quality inspection through exploratory data analysis, statistical description, regression modelling, and evaluation — on the Student Performance Factors dataset. Attendance and study hours showed the strongest numerical relationships with Exam_Score, while parental involvement, resource access, and peer influence showed smaller but consistent group-level differences. Among the three regression models compared, Linear Regression achieved the best and most consistent generalisation on the held-out test set (test RMSE = 1.521, MAE = 0.416, R^2 = 0.711), and was

therefore selected as the final model. The project demonstrates that easily observable behavioural indicators — attendance in particular — carry meaningful predictive value for exam performance in this dataset.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

- Monitor attendance as a practical, low-cost indicator for identifying students who may benefit from early academic support, since it showed the strongest association with Exam_Score.
- Encourage structured, consistent study routines, as study effort was positively and substantially associated with exam performance.
- Investigate whether students with lower access to educational resources require additional institutional support, given the observed group-level score gap.
- Explore positive peer-learning environments (e.g. study groups) in future interventions, informed by the observed peer-influence differences.
- Validate the selected model on new, out-of-sample data (e.g. a different academic term or institution) before using it to inform real-world decisions, given the limitations noted in Section 15.

17.2 Future Work

Future extensions of this project could include: testing regularised linear models (e.g. Ridge or Lasso) alongside more carefully tuned tree-based ensembles to control the overfitting observed in Random Forest; incorporating additional socio-economic or well-being variables if available; validating the model against data from other cohorts or academic years to test generalisation; and exploring whether a classification framing (e.g. pass/fail or performance bands) offers additional decision-support value alongside the continuous Exam_Score prediction.

18. REFERENCES

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APPENDIX A. CODE

This appendix presents illustrative code extracts corresponding to each stage of the analysis. The complete, executable workflow is provided in the accompanying Jupyter/Colab notebook.

A.1 Data Loading and Inspection

```
import pandas as pd

df = pd.read_csv('P_2_StudentPerformanceFactors.csv')
print(df.shape)
print(df.dtypes)
```

```
print(df.isnull().sum())
print(df.duplicated().sum())
```

A.2 Data Cleaning

```
# Remove invalid Exam_Score values outside the 0-100 range
df = df[df['Exam_Score'] <= 100].copy()
print(df.shape) # (6606, 20)
```

A.3 Preprocessing

```
numeric_cols = ['Hours_Studied', 'Attendance', 'Sleep_Hours',
                'Previous_Scores', 'Tutoring_Sessions', 'Physical_Activity']
categorical_cols = ['Parental_Involvement', 'Access_to_Resources',
                   'Extracurricular_Activities', 'Motivation_Level', 'Internet_Access',
                   'Family_Income', 'Teacher_Quality', 'School_Type', 'Peer_Influence',
                   'Learning_Disabilities', 'Parental_Education_Level',
                   'Distance_from_Home', 'Gender']
X = df[numeric_cols + categorical_cols]
y = df['Exam_Score']
```

A.4 EDA and Visualization

```
import matplotlib.pyplot as plt

plt.hist(df['Exam_Score'], bins=15, edgecolor='black')
plt.title('Distribution of Exam Scores')
plt.xlabel('Exam Score'); plt.ylabel('Number of Students')
plt.show()

df[numeric_cols].corrwith(df['Exam_Score']).sort_values().plot(kind='barh')
```

A.5 Model Training and Evaluation

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

for name, mdl in [('Linear Regression', linreg_model),
                  ('Random Forest', rf_model),
                  ('Gradient Boosting', gb_model)]:
    preds = mdl.predict(X_test)
    rmse = np.sqrt(mean_squared_error(y_test, preds))
    mae = mean_absolute_error(y_test, preds)
    r2 = r2_score(y_test, preds)
    print(name, rmse, mae, r2)
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

All figures and tables relevant to this analysis are presented within the main report (Sections 8, 9, 12 and 13). No additional supplementary figures were required beyond those already included in the main text.